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Sheet post-processing device and image forming system including same

Abstract

A sheet post-processing device includes a conveying member, a processing tray, a processing section, a reference plate, an alignment member, and a frame. The processing tray stacks sheets carried in by the conveying member. The alignment member moves the sheet carried in onto the processing tray in an alignment direction so as to aid alignment of the sheets. The alignment member includes a roller holder, an alignment roller, and a guide member. The guide member has a sheet guide part protruding from a swinging end of the roller holder to the downstream side in the alignment direction than the alignment roller. When the number of the sheets placed on the processing tray varies, an angle of the guide member with respect to the roller holder varies so that a distance between a top sheet placed on the processing tray and the sheet guide part becomes substantially constant.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7976004	12/2010	Kimura	270/58.11	B65H 31/3027
8123209	12/2011	Kubota	270/58.08	B65H 31/36
8172214	12/2011	Kubota	270/58.11	B65H 31/34
8439342	12/2012	Okamoto	270/58.11	B65H 39/00
8602405	12/2012	Okamoto	270/58.11	B65H 39/00
8874022	12/2013	Kayaba	271/220	B65H 29/6618
9016681	12/2014	Mizubata	270/58.11	G03G 15/6552
10941013	12/2020	Egawa	N/A	B65H 31/02
2006/0208411	12/2005	Saito et al.	270/58.07	N/A
2020/0071116	12/2019	Egawa	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2006-256729	12/2005	JP	N/A
2020-033156	12/2019	JP	N/A

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-000548 filed Jan. 5, 2023, the entire contents of which are hereby incorporated by reference.

BACKGROUND

(2) The present disclosure relates to a sheet post-processing device that performs predetermined

post-processing on a sheet after image formation by an image forming apparatus, and to an image forming system including the same.

(3) There is known a sheet post-processing device, which stacks a plurality of sheets after image formation by the image forming apparatus such as a copier or a printer, and performs a stapling process in which the stacked sheets are stapled as a whole by a stapler, a hole punching process in which punch holes (perforation holes) are formed in the sheets by a hole punching device, or a folding process in which a fold line is formed in the sheets.

(4) This sheet post-processing device is equipped with a processing tray on which a predetermined number of sheets after image formation are placed. Then, the stapling process, a shift discharge process (sorting process), or the like is performed on the sheets placed on the processing tray. In addition, in order to smoothly perform the stapling process, the shift discharge process, or the like, the sheets on the processing tray are aligned using an alignment member such as an alignment roller.

SUMMARY

(5) A sheet post-processing device according to one aspect of the present disclosure includes a conveying member, a processing tray, a processing section, a reference plate, an alignment member, and a frame. The conveying member conveys a sheet. The processing tray stacks a plurality of sheets conveyed in a predetermined carry-in direction by the conveying member. The processing section performs predetermined post-processing on the sheets placed on the processing tray. The reference plate abuts an edge of the sheet carried in onto the processing tray on a downstream side in an alignment direction opposite to the carry-in direction, so as to align the sheet in the carry-in direction. The alignment member moves the sheet carried in onto the processing tray in the alignment direction so as to aid alignment of the sheets. The frame supports the conveying member and the alignment member. The alignment member includes a roller holder, an alignment roller, and a guide member. The roller holder is disposed above the processing tray, and is supported by the frame in a swingable manner in the carry-in direction. The alignment roller is supported in a rotatable manner at a swinging end of the roller holder so as to contact with a top sheet placed on the processing tray. The guide member has a sheet guide part that protrudes toward the downstream side in the carry-in direction from the swinging end of the roller holder more than the alignment roller. The guide member is supported in a swingable manner at the swinging end of the roller holder. When the number of sheets placed on the processing tray varies, an angle of the guide member with respect to the roller holder varies so that a distance between the top sheet placed on the processing tray and the sheet guide part will be substantially constant.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram illustrating a structure of an image forming system constituted of a sheet post-processing device according to an embodiment of the present disclosure, and an image forming apparatus to which the sheet post-processing device is connected.

(2) FIG. 2 is a cross-sectional side view schematically illustrating a structure of the sheet post-processing device of this embodiment.

(3) FIG. 3 is a perspective view of a sheet stapling unit mounted on the sheet post-processing device of this embodiment.

(4) FIG. 4 is a side view of the sheet stapling unit.

(5) FIG. 5 is a cross-sectional side view illustrating a processing tray, an alignment member, and their vicinity.

(6) FIG. 6 is a cross-sectional side view illustrating the processing tray, the alignment member, and their vicinity, in a state where no sheet is carried in onto the processing tray

- (7) FIG. 7 is a diagram illustrating a state where a sheet is carried in onto the processing tray from the state illustrated in FIG. 6.
- (8) FIG. 8 is a diagram illustrating a state where another sheet is carried in onto the processing tray from the state illustrated in FIG. 7.

DETAILED DESCRIPTION

- (9) Hereinafter, with reference to the drawings, an embodiment of the present disclosure is described in detail. FIG. 1 is a schematic diagram illustrating a structure of an image forming system constituted of a sheet post-processing device 1 including a sheet stapling unit 92 according to an embodiment of the present disclosure, and an image forming apparatus 200 to which the sheet post-processing device 1 is connected.
- (10) As illustrated in FIG. 1, the image forming apparatus 200 prints an image on a sheet (paper sheet), on the basis of image data input externally via a not-shown network communication unit, or image data read by an image reading unit 201 disposed at an upper part of the image forming apparatus 200.
- (11) The sheet post-processing device 1 is connected to a side face of the image forming apparatus 200 in an attachable and detachable manner. The sheet post-processing device 1 performs post-processing such as a hole punching process, a stapling process, or the like, on sheets after image formation (printing) by the image forming apparatus 200. Note that the sheet post-processing device 1 is not limited to one that performs the post-processing on the sheets conveyed automatically from the image forming apparatus 200, but may be one that conveys sheets set on a not-shown tray by a user to a position at which the post-processing can be performed, and performs the post-processing on the sheets.
- (12) FIG. 2 is a cross-sectional side view schematically illustrating a structure of the sheet post-processing device 1 of this embodiment. As illustrated in FIG. 2, the sheet post-processing device 1 includes a sheet carry-in entrance 2, a first sheet conveying path 3, a first sheet discharge unit 4, a second sheet conveying path 5, a second sheet discharge unit 6, a third sheet conveying path 7, a third sheet discharge unit 8, a post-processing unit 9, and a post-processing control unit (control unit) 10.
- (13) The sheet carry-in entrance 2 is an opening formed on the side face of the sheet post-processing device 1, which faces the image forming apparatus 200. The sheet conveyed from the image forming apparatus 200 to the sheet post-processing device 1 passes through the sheet carry-in entrance 2 and is carried into the sheet post-processing device 1.
- (14) The first sheet conveying path 3 extends in a substantially horizontal direction from the sheet carry-in entrance 2 to the first sheet discharge unit 4, in a direction separating from the image forming apparatus 200 (to the left direction in FIG. 2). Note that the direction from the sheet carry-in entrance 2 to the first sheet discharge unit 4 is referred to as a sheet conveying direction of the first sheet conveying path 3. The sheet carry-in entrance 2 is positioned at an upstream end of the first sheet conveying path 3 in the sheet conveying direction. The first sheet conveying path 3 includes a plurality of convey roller pairs 3r, and conveys the sheet carried in from the sheet carry-in entrance 2 to the sheet post-processing device 1, toward the downstream side in the sheet conveying direction.
- (15) The first sheet discharge unit 4 is disposed at a side face of the sheet post-processing device 1, which is opposite to the side face facing the image forming apparatus 200. The first sheet discharge unit 4 is disposed at a downstream end of the first sheet conveying path 3 in the sheet conveying direction. The first sheet discharge unit 4 includes a first discharge outlet 41, a first discharge roller pair 42, and a first discharge tray 43.
- (16) The first discharge outlet 41 is positioned at the downstream end of the first sheet conveying path 3 in the sheet conveying direction. The first discharge roller pair 42 is disposed at the first discharge outlet 41. The first discharge tray 43 is positioned on the downstream side of the first discharge outlet 41 in the sheet conveying direction. The sheet, which is conveyed along the first

sheet conveying path **3** and reaches the first discharge outlet **41**, is discharged onto the first discharge tray **43** by the first discharge roller pair **42** after passing through the first discharge outlet **41**. The first discharge tray **43** is one of final discharge places of the sheets after post-processing by the sheet post-processing device **1**.

(17) The second sheet conveying path **5** branches from the first sheet conveying path **3** at a first junction (junction) **31** and extends laterally and upwardly to the second sheet discharge unit **6**, in the direction separating from the image forming apparatus **200** (to the left direction in FIG. **2**). The first junction **31** is disposed on the downstream side of a perforation unit **91** in the sheet conveying direction of the first sheet conveying path **3**. Note that the direction from the first junction **31** to the second sheet discharge unit **6** is referred to as a sheet conveying direction of the second sheet conveying path **5**. The first junction **31** is positioned at the upstream end of the second sheet conveying path **5** in the sheet conveying direction. The second sheet conveying path **5** includes a plurality of convey roller pairs **5r**, and allows the sheet conveyed along the first sheet conveying path **3** to branch at the first junction **31**, and conveys the sheet to the second sheet discharge unit **6**.

(18) The first junction **31** includes a first switching guide **311**. The first switching guide **311** swings between a position for guiding the sheet, which is conveyed from the sheet carry-in entrance **2** along the first sheet conveying path **3**, to the first discharge outlet **41** along the first sheet conveying path **3**, and a position for guiding the sheet to the second sheet conveying path **5** after branching from the first sheet conveying path **3**. Furthermore, the first switching guide **311** swings to a position for guiding the sheet, which passes through a later-described second folding conveying path **106** after a folding process, to the second sheet conveying path **5**. The first switching guide **311** is connected to a drive mechanism (not shown), and its operation is controlled by a post-processing control unit **10**.

(19) The second sheet discharge unit **6** is disposed above the first sheet discharge unit **4** on the side face of the sheet post-processing device **1**, which is opposite to the side face facing the image forming apparatus **200**. The second sheet discharge unit **6** is disposed at the downstream end of the second sheet conveying path **5** in the sheet conveying direction. The second sheet discharge unit **6** includes a second discharge outlet **61**, a second discharge roller pair **62**, and a second discharge tray **63**.

(20) The second discharge outlet **61** is positioned at the downstream end of the second sheet conveying path **5** in the sheet conveying direction. The second discharge roller pair **62** is disposed at the second discharge outlet **61**. The second discharge tray **63** is positioned on the downstream side of the second discharge outlet **61** in the sheet conveying direction. The sheet, which is conveyed along the second sheet conveying path **5** and reaches the second discharge outlet **61**, is discharged onto the second discharge tray **63** by the second discharge roller pair **62** after passing through the second discharge outlet **61**. The second discharge tray **63** is one of the final discharge places of the sheets after post-processing by the sheet post-processing device **1**. In addition, sheets on which the post-processing is not performed, small size sheets, or the like are also discharged onto the second discharge tray **63**.

(21) The third sheet conveying path **7** branches from the first sheet conveying path **3** at a second junction **32**, and extends downward to the third sheet discharge unit **8**. Note that the direction from the second junction **32** to the third sheet discharge unit **8** is referred to as a sheet conveying direction of the third sheet conveying path **7**. The second junction **32** is positioned on the downstream side of the first junction **31** in the sheet conveying direction of the first sheet conveying path **3**, and is positioned at the upstream end of the third sheet conveying path **7** in the sheet conveying direction. The third sheet conveying path **7** includes a plurality of convey roller pairs **7r**, allows the sheet conveyed along the first sheet conveying path **3** to branch at the second junction **32**, and conveys the same to the third sheet discharge unit **8**.

(22) The second junction **32** includes a second switching guide **321**. The second switching guide **321** swings between a position for guiding the sheet, which is conveyed from the sheet carry-in

entrance **2** along the first sheet conveying path **3**, to the first discharge outlet **41** along the first sheet conveying path **3**, and a position for guiding the sheet, which is conveyed from the sheet carry-in entrance **2** along the first sheet conveying path **3** and is switched back after passing through the second junction **32**, to the third sheet conveying path **7**. The second switching guide **321** is connected to a drive mechanism (not shown), and its operation is controlled by the post-processing control unit **10**.

(23) The third sheet discharge unit **8** is disposed below the first sheet discharge unit **4** (near a lower end part of the sheet post-processing device **1**), on the side face of the sheet post-processing device **1**, which is opposite to the side face facing the image forming apparatus **200**. The third sheet discharge unit **8** includes a third discharge outlet **81**, a third discharge roller pair **82**, and a third discharge tray **83**.

(24) The third discharge outlet **81** is positioned at the downstream end of the third sheet conveying path **7** in the sheet conveying direction. The third discharge roller pair **82** is disposed at the third discharge outlet **81**. The third discharge tray **83** is positioned on the downstream side of the third discharge outlet **81** in the sheet conveying direction. The sheet, which is conveyed along the third sheet conveying path **7** and reaches the third discharge outlet **81**, is discharged onto the third discharge tray **83** by the third discharge roller pair **82** after passing through the third discharge outlet **81**. The third discharge tray **83** is one of the final discharge places of the sheets after post-processing by the sheet post-processing device **1**.

(25) The post-processing unit **9** performs predetermined post-processing on the sheets carried into the sheet post-processing device **1** after image formation by the image forming apparatus **200**. The post-processing unit **9** includes the perforation unit **91**, the sheet stapling unit **92**, a sheet folding unit **100**, and a binding unit **94**.

(26) The perforation unit **91** is disposed near the downstream side of the sheet carry-in entrance **2** of the first sheet conveying path **3**. The perforation unit **91** performs a perforation process on the sheet conveyed along the first sheet conveying path **3**, so as to form punch holes.

(27) The sheet stapling unit **92** is disposed near the upstream side of the first sheet discharge unit **4** in the sheet conveying direction of the first sheet conveying path **3**. The sheet stapling unit **92** performs the stapling process on a bundle of sheets formed by stacking a plurality of sheets, so as to staple the sheets. A detailed structure of the sheet stapling unit **92** will be described later.

(28) The sheet folding unit **100** is disposed on the downstream side of the perforation unit **91** and the upstream side of the stapling unit **92**, in the sheet conveying direction of the first sheet conveying path **3**. The sheet folding unit **100** performs the folding process on one sheet so as to form a fold line. The sheet folding unit **100** can perform the folding process on one sheet, such as half-fold, Z-fold, outer three-fold, inner three-fold, or the like.

(29) The binding unit **94** is disposed near the upstream side of the third sheet discharge unit **8** in the sheet conveying direction of the third sheet conveying path **7**. The binding unit **94** includes a center-folding unit **941** and a center-stapling unit **942**. The binding unit **94** performs a center-folding process and a center-stapling process for folding and stapling the substantial center of a bundle of sheets formed by stacking a plurality of sheets, in the sheet conveying direction, so as to make a booklet.

(30) The post-processing control unit **10** includes a CPU, a storage unit, and other electronic circuits and components (which are not shown). The post-processing control unit **10** is connected to a main body control unit of the image forming apparatus **200** (see FIG. **1**) in a communicable manner. The post-processing control unit **10** receives a command from the main body control unit, controls operations of individual components of the sheet post-processing device **1**, and performs processing related to functions of the sheet post-processing device **1**, using the CPU on the basis of control programs and data stored in the storage unit. The first sheet conveying path **3**, the first sheet discharge unit **4**, the second sheet conveying path **5**, the second sheet discharge unit **6**, the third sheet conveying path **7**, the third sheet discharge unit **8**, and the post-processing unit **9** each receive

commands from the post-processing control unit **10** individually, and cooperate to perform post-processing on the sheet. Note that it may be possible that functions of the post-processing control unit (control unit) **10** are performed by the main body control unit of the image forming apparatus **200**.

(31) Next, a structure of the sheet stapling unit **92** is described. FIG. **3** is a perspective view of the sheet stapling unit **92** mounted on the sheet post-processing device **1**. FIG. **4** is a side view of the sheet stapling unit **92**.

(32) As illustrated in FIG. **3**, the sheet stapling unit **92** includes a processing tray **521**, a stapling unit **71**, and a reference plate **73**.

(33) The processing tray **521** is a rectangular tray that extends in a sheet width direction (arrow AA' direction) and in a carry-in direction. A plurality of sheets S (the bundle of sheets) to be stapled are placed on the processing tray **521**. In this case, the sheets S are carried in onto the processing tray **521** in the alignment direction toward the lower right direction in FIG. **4** (arrow B direction, opposite direction to the carry-in direction). The bundle of sheets after the stapling process is eventually sent out by the first discharge roller pair **42** (see FIG. **2**) in the opposite direction to the above alignment direction (to the upper left direction in FIG. **4**), and is discharged onto the first discharge tray **43** (see FIG. **2**). The downstream side of the processing tray **521** in the carry-in direction (the lower left side in FIG. **3**) supports a lower side roller **421** constituting the first discharge roller pair **42**.

(34) The processing tray **521** includes a tray middle part **522** and width restricting members **523**. The tray middle part **522** is disposed on the top surface part of the processing tray **521** in the middle part in the sheet width direction. The tray middle part **522** is a sheet-like member fixed on the processing tray **521** with a slight height.

(35) The width restricting members **523** are disposed in pair on both sides of the tray middle part **522** in the sheet width direction. The width restricting members **523** restrict a position in the sheet width direction of the sheet S carried in onto the processing tray **521**. The width restricting member **523** is constituted of a sheet-like member similarly to the tray middle part **522**, and has a sidewall standing upward at an end in the sheet width direction. The processing tray **521** is provided with a guide groove **524** formed to extend in the sheet width direction. The width restricting member **523** can be moved in a reciprocating manner along the guide groove **524** in the sheet width direction, by a drive mechanism such as a rack and pinion gear (not shown). In this embodiment, every time when the sheet is carried in onto the processing tray **521**, the width restricting members **523** are driven by the drive mechanism to move in a reciprocating manner. As a result, the sheet placed on the processing tray **521** is aligned in the sheet width direction.

(36) The stapling unit **71** is disposed so as to face an end edge of the sheet on the front end side in the alignment direction (the right side in FIG. **4**). The stapling unit **71** can be moved by drive force of a motor M along the end edge of the sheet in the sheet width direction (arrow AA' direction) perpendicular to the carry-in direction, and performs the stapling process on the bundle of the sheets.

(37) As illustrated in FIG. **4**, the stapling unit **71** includes a staple main body part **711** and a staple movable part **712**. The staple main body part **711** is a main body part of the stapling unit **71**, in which a plurality of staple needles (not shown) are housed. The staple movable part **712** can move in the up and down direction, so as to insert the staple needle into the sheets S. Between the staple main body part **711** and the staple movable part **712**, there is formed a recess **713** into which the end edge of the sheet enters.

(38) The reference plates **73** are fixed at three positions separated in the sheet width direction, so as to face the end of the tray middle part **522** on the downstream side (the upper right side in FIG. **3** and the lower right side in FIG. **4**) in the alignment direction. The reference plate **73** has a substantial U-shape that opens to the upstream side (the upper left side in FIG. **4**) in the alignment direction, in a cross section perpendicular to the sheet width direction. The reference plate **73** abuts

the end edge of the sheet carried in onto the processing tray 521, so as to align the sheet in the carry-in direction.

(39) FIG. 5 is a cross-sectional side view illustrating a structure of the processing tray 521, an alignment member 55, and their vicinity. As illustrated in FIG. 5, a carry-in roller pair 54 is provided to a conveyance frame 53 positioned above the processing tray 521. The carry-in roller pair 54 is constituted of an upper carry-in roller 54a and a lower carry-in roller 54b.

(40) A sheet detection sensor 93 is disposed in a vicinity of the carry-in roller pair 54. The sheet detection sensor 93 detects timing when the sheet S passes the carry-in roller pair 54. As the sheet detection sensor 93, for example, a photo interrupter (PI) sensor is used, which includes a detection unit constituted of a light emitting part and a light receiving part.

(41) The alignment member 55 is disposed below the carry-in roller pair 54. The alignment member 55 moves the sheet S carried in onto the processing tray 521 in the alignment direction to approach the reference plate 73, so as to aid the alignment of the sheet S. The alignment member 55 includes a roller holder 56, an alignment roller 57, and a guide member 58.

(42) The roller holder 56 is supported by the conveyance frame 53 at the lower end part in a swingable manner along the carry-in direction of the sheet S. The alignment roller 57 is supported by the roller holder 56 at the swinging end in a rotatable manner. The alignment roller 57 is supplied with a rotation drive force from a drive source such as a motor (not shown), in a direction to send out the sheet S in the alignment direction (counterclockwise direction in FIG. 5). When the alignment roller 57 contacts with the upper surface of the sheet S carried in onto the processing tray 521, the sheet S is moved in the alignment direction (arrow B direction), and the end edge of the sheet S abuts the reference plate 73 for alignment.

(43) The rotation drive of the alignment roller 57 is controlled based on detection timing of the sheet detection sensor 93. Specifically, the time is calculated in advance, which is from the timing when the sheet detection sensor 93 detects passing of the sheet S until the sheet S is carried in onto the processing tray 521 and is aligned by the reference plate 73. Then, the rotation drive of the alignment roller 57 is stopped at the timing when the above time elapses from the detection timing of the sheet S.

(44) Alternatively, a roller having a small friction coefficient with the sheet S is used as the alignment roller 57, so that rotation of the alignment roller 57 may be continued after the sheet S is aligned by the reference plate 73. With this structure, after the sheet S abuts the reference plate 73, the alignment roller 57 idles (slips). Therefore, even if the friction coefficient of the sheet S varies due to a surface state of the sheet S or a moisture absorption state of the same, the sheet S can be securely aligned by abutting the reference plate 73.

(45) The guide member 58 includes a fulcrum part 58a supported by a rotation shaft 57a of the alignment roller 57 in a swingable manner, a sheet guide part 58b extending from the fulcrum part 58a to the downstream side in the alignment direction of the sheet S (the lower right side in FIG. 5), and an angle adjusting part 58c extending to the upstream side in the alignment direction (the upper left side in FIG. 5). The sheet guide part 58b is disposed to face the sheet S carried in onto the processing tray 521 near the end edge on the downstream side in the alignment direction, so as to suppress uplift or bulge of the sheet S.

(46) An arm member 59 is supported by a rotation shaft 541 of the lower carry-in roller 54b in a swingable manner. The arm member 59 receives a rotation drive force from a drive source such as a motor (not shown), in a forward direction (the counterclockwise direction in FIG. 5) and in a reverse direction (the clockwise direction in FIG. 5). An engagement boss 56a is formed on a side face of the roller holder 56. When the arm member 59 is swung in the forward direction, the arm member 59 contacts with the engagement boss 56a from below. When the arm member 59 is swung in the reverse direction, the arm member 59 separates from the engagement boss 56a.

(47) The swinging of the arm member 59 is controlled based on the detection timing of the sheet detection sensor 93. Specifically, at the timing when the sheet detection sensor 93 detects passing

of the sheet S, the arm member 59 is swung in the forward direction, and hence the arm member 59 contacts with the engagement boss 56a from below and lifts up the roller holder 56. As a result, the alignment roller 57 separates from the top surface of the processing tray 521 (or the sheet S placed on the processing tray 521).

(48) Further, at the timing when the end edge of a new sheet carried in onto the processing tray 521 passes under the alignment roller 57, the arm member 59 is swung in the reverse direction. As a result, the arm member 59 separates from the engagement boss 56a, and hence the roller holder 56 swings downward by its weight, so that the alignment roller 57 contacts with the top surface of the sheet S. The operation described above is performed every time when the sheet S is carried in, and hence the alignment roller 57 can securely contact with the top surface of the sheet S, while avoiding interference between the alignment roller 57 and the end edge of the sheet carried in onto the processing tray 521.

(49) The arm member 59 is provided with an arc-shaped sliding surface 59a formed near the rotation shaft 541. When the roller holder 56 swings upward (in the clockwise direction in FIG. 5), the angle adjusting part 58c of the guide member 58 contacts with the sliding surface 59a.

(50) Next, there is described an operation of the alignment member 55 when the number of the sheets S carried in onto the processing tray 521 is increased. FIG. 6 is a cross-sectional side view illustrating a structure of the processing tray 521, the alignment member 55, and their vicinity, in a state where no sheet S is carried in onto the processing tray 521.

(51) If no sheet S is carried in onto the processing tray 521, as illustrated in FIG. 6, the roller holder 56 is swung downward (in the counterclockwise direction in FIG. 6) by gravity acting on the roller holder 56 and the alignment roller 57, and the alignment roller 57 is contacted with the top surface of the processing tray 521.

(52) The center of gravity of the guide member 58 is positioned closer to the sheet guide part 58b than the fulcrum part 58a. Therefore, in a state where no external force is applied to the guide member 58, the guide member 58 tends to swing in the direction for the sheet guide part 58b to descend by its weight (in the clockwise direction in FIG. 6). In addition, the roller holder 56 is provided with a restricting part (not shown) that restricts the swinging of the guide member 58 by its weight at a predetermined position. Therefore, as illustrated in FIG. 6, the guide member 58 is supported in a position where the sheet guide part 58b is substantially parallel to the processing tray 521. In this state, the angle adjusting part 58c of the guide member 58 is apart from the sliding surface 59a.

(53) When the carry-in of the sheet S onto the processing tray 521 is started, the sheet S carried into the sheet post-processing device 1 from the sheet carry-in entrance 2 (see FIG. 2) passes through the first sheet conveying path 3 (see FIG. 2) and is carried in onto the processing tray 521 by the carry-in roller pair 54. In this case, the first discharge roller pair 42 is in a separate state (a nip released state), and the front end of the sheet S passes through the first discharge roller pair 42 and protrudes above the first discharge tray 43 (see FIG. 2). In addition, when the arm member 59 is swung in the forward direction, the roller holder 56 is lift up, and the alignment roller 57 is apart from the top surface of the processing tray 521.

(54) At the timing when the rear end of the sheet S passes the carry-in roller pair 54, the post-processing control unit 10 (see FIG. 2) controls the arm member 59 to swing in the reverse direction. In this way, the roller holder 56 swing downward, and the alignment roller 57 contacts with the top surface of the sheet S. The sheet S is sent to the downstream side in the alignment direction by the alignment roller 57, so as to be aligned in the sheet width direction by the width restricting member 523 (see FIG. 3), and be aligned in the carry-in direction by the reference plate 73, and is stacked.

(55) After a predetermined number of the sheets S are carried in onto the processing tray 521 and are aligned by the reference plate 73, the post-processing control unit 10 (see FIG. 2) controls the stapling unit 71 to perform the stapling process, controls the first discharge roller pair 42 to become

a contact state (a nip formed state), and controls the first discharge roller pair **42** to rotate in a discharge direction. In this way, the bundle of sheets **S** is discharged onto the first discharge tray **43**.

(56) FIG. **7** is a diagram illustrating a state where the sheet **S** is carried in onto the processing tray **521** from the state illustrated in FIG. **6**. When the sheet **S** is carried in onto the processing tray **521**, the alignment roller **57** is lift up by the thickness of the sheet **S**. As a result, the roller holder **56** is also swung upward (in the clockwise direction in FIG. **7**). When the roller holder **56** is swung upward, the angle adjusting part **58c** of the guide member **58** contacts with the lower end part of the sliding surface **59a**.

(57) In this way, the guide member **58** slightly swings in the counterclockwise direction from the state illustrated in FIG. **6**, and the sheet guide part **58b** moves in the direction to separate from a top sheet **S1**. As a result, the guide member **58** is retained in a position where a predetermined distance **d** is secured between the sheet guide part **58b** and the top sheet **S1**.

(58) FIG. **8** is a diagram illustrating a state where another sheet **S** is carried in onto the processing tray **521** from the state illustrated in FIG. **7**. When another sheet **S** is carried in onto the processing tray **521**, the alignment roller **57** is further lift up by the thickness of the sheet **S**. As a result, the roller holder **56** also further swings upward (in the clockwise direction in FIG. **8**).

(59) When the roller holder **56** swings upward, the angle adjusting part **58c** of the guide member **58** moves upward along the sliding surface **59a**. In this case, the angle adjusting part **58c** contacts with the sliding surface **59a** and moves slightly in the direction to approach the processing tray **521** (in the clockwise direction in FIG. **8**), and hence the sheet guide part **58b** moves in the direction to separate from the processing tray **521**.

(60) The shape of the sliding surface **59a** is designed so that a movement of the angle adjusting part **58c** (the sheet guide part **58b**) varies in accordance with variation of the upward swing angle of the roller holder **56**, and that the distance **d** between the sheet guide part **58b** and the top sheet **S1** is maintained to be constant. Note that it may be possible that when the arm member **59** swings, an angle of the sliding surface **59a** with respect to the top surface of the processing tray **521** varies, so that the distance **d** between the sheet guide part **58b** and the top sheet **S1** is maintained to be constant.

(61) According to the structure described above, the guide member **58** swings in accordance with the number of the sheets **S** placed on the processing tray **521**, and the distance **d** between the sheet guide part **58b** and the top sheet **S1** is maintained. Therefore, there is no risk that the sheet guide part **58b** will contact with the top sheet **S1** so as to disturb the alignment operation, or that the sheet guide part **58b** will separate from the top sheet **S1** so as to cause a variation in the effect of suppressing bulge of the sheet **S**. Therefore, the sheets **S** can be securely aligned in the state where a small number of the sheets **S** are placed on the processing tray **521**, and in the state where the maximum allowable number of the sheets **S** are placed on the same.

(62) In addition, when the roller holder **56** swings, the angle adjusting part **58c** slides along the arc-shaped sliding surface **59a** formed on the arm member **59**, and hence the angle of the guide member **58** with respect to the roller holder **56** is changed. Therefore, without a complicated structure, the distance **d** between the sheet guide part **58b** and the top sheet **S1** can be maintained to be constant with a simple structure.

(63) Although the embodiment of the present disclosure is described above, the scope of the present disclosure is not limited to this, but can be variously modified for implementation within the scope of the invention without deviating from the spirit thereof. For instance, in the embodiment described above, the case where the sheet stapling unit **92** performs the stapling process on the sheets **S** stacked on the processing tray **521** is exemplified, but without limiting to this, the present disclosure may have a structure in which a shift discharge process or the folding process is performed on the sheets **S** stacked on the processing tray **521**.

(64) In addition, in the embodiment described above, the sliding surface **59a** is formed integrally to

the arm member 59 in the vicinity of the swing fulcrum, but it may be possible to dispose a member having the sliding surface 59a separately from the arm member 59.

(65) The present disclosure can be used for a sheet post-processing device that performs predetermined post-processing on a plurality of sheets.

Claims

1. A sheet post-processing device comprising: a conveying member configured to convey a sheet; a processing tray configured to stack a plurality of the sheets carried in by the conveying member in a predetermined carry-in direction; a processing section configured to perform predetermined post-processing on the sheets placed on the processing tray; a reference plate configured to abut an end edge of the sheet carried in onto the processing tray, on a downstream side in an alignment direction opposite to the carry-in direction, so as to align the sheet in the carry-in direction; an alignment member configured to move the sheet carried in onto the processing tray in the alignment direction so as to aid alignment of the sheets; and a frame for supporting the conveying member and the alignment member, wherein the alignment member includes a roller holder disposed above the processing tray so as to be supported by the frame in a swingable manner in the carry-in direction, an alignment roller supported in a rotatable manner at a swinging end of the roller holder so as to contact with a top sheet placed on the processing tray, and a guide member having a sheet guide part protruding toward the downstream side in the alignment direction from the swinging end of the roller holder more than the alignment roller, the sheet guide part facing a top sheet among the sheets stacked on the processing tray across a predetermined distance therefrom, and the guide member is supported in a swingable manner at the swinging end of the roller holder, and when the number of the sheets placed on the processing tray varies, an angle of the guide member with respect to the roller holder varies so that a distance between the top sheet placed on the processing tray and the sheet guide part becomes substantially constant, the guide member further includes a fulcrum part supported in a swingable manner at the swinging end of the roller holder, and an angle adjusting part protruding from the fulcrum part to the upstream side in the alignment direction, the sheet guide part extends from the fulcrum part in the alignment direction, a sliding surface is provided, which is disposed to face the angle adjusting part and contacts with the angle adjusting part when the roller holder swings, so as to restrict the sheet guide part from swinging downward, and when the roller holder swings, the angle adjusting part slides along the sliding surface so that the angle of the guide member with respect to the roller holder varies.
2. The sheet post-processing device according to claim 1, further comprising a control unit configured to control swinging of the roller holder, wherein the control unit controls the roller holder to swing upward when the number of the sheets placed on the processing tray increases, and when the roller holder swings upward, the guide member swings about the fulcrum part in the direction where the sheet guide part separates from the processing tray.
3. The sheet post-processing device according to claim 1, further comprising an arm member whose one end is supported by the frame in a swingable manner, the arm member being configured to allow the roller holder to swing in the direction to separate from the processing tray when the sheet is carried in onto the processing tray, wherein the sliding surface is formed on one end of the arm member, and an angle of the sliding surface with respect to the top surface of the processing tray varies when the arm member swigs.
4. The sheet post-processing device according to claim 1, wherein the processing section is a stapling unit configured to perform a stapling process to staple the plurality of sheets placed on the processing tray with a staple needle at a predetermined position.
5. An image forming system comprising: an image forming apparatus configured to form an image

on a sheet; and the sheet post-processing device according to claim 1, configured to perform a predetermined post-processing on the sheet after image formation by the image forming apparatus.
